

An exact third-denominator return between the original Erdős–Straus channels

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Abstract

An original rational middle trace with denominator at most three always returns to an integer exterior or middle state at the same shell when $\gcd(R, p-1) = R/3$. In fact the entire original shell count is exactly one half of the third-denominator trace count. The proof uses the original complementary divisor and an explicit affine residue, without numerical estimation or an assumed integer witness. At $R = 27$ this gives a complete factor-level classification and a constructive return using at most three signed prime occurrences. The initial-source existence assertion for an arbitrary hard prime remains unproved. This is a local existence lemma and an exact exceptional-locus theorem, not universal Erdős–Straus coverage.

1 Original domain and markings

Let $p \equiv 1 \pmod{24}$ be prime and let

$$0 < R < p, \quad R \equiv 3 \pmod{4}, \quad a = (p + R)/4.$$

In particular $p/4 < a < p/2$ and $\gcd(a, pR) = 1$. Retain the entire divisor box

$$a = \prod_q q^{e_q}, \quad u = \prod_q q^{f_q}, \quad 0 \leq f_q \leq 2e_q.$$

Define the two original integer gates

$$E = \{u \mid a^2 : R \mid 4u + 1\}, \quad M = \{u \mid a^2 : R \mid p + 4u\}.$$

The normalization and original ordered reconstructions are

$$\begin{aligned} g &= \gcd(a, u), & (h, r, s) &= (g^2/u, u/g, a/g), \\ E : \quad \kappa &= (pr + s)/R, & (x, y, z) &= (a, h\kappa, phr\kappa), \\ M : \quad \lambda &= (r + s)/R, & (x, y, z) &= (a, phs\lambda, phr\lambda). \end{aligned}$$

Here $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$. The identities are valid over the positive rationals before gate integrality is imposed. For example the middle tails are

$$y = \frac{p(a + a^2/u)}{R}, \quad z = \frac{p(a + u)}{R},$$

and $1/a + 1/y + 1/z = 4/p$ by direct multiplication. Their ordered inverse is

$$u = \frac{a^2}{Ry - pa} \quad (E), \quad u = \frac{pa^2}{Ry - pa} \quad (M).$$

These are the classical Type-I/II coordinates, with the specific marking fixed here; see Elsholtz and Tao (2013), Section 2, Propositions 2.2 and 2.6. All returns below keep p, R, a and either keep u or replace it by its actual complementary divisor a^2/u . No root, prime factor, or exponent outside the original box is introduced.

2 Complete same-shell cancellation at denominator three

Theorem 1 (Third-denominator return and count). *Assume in addition*

$$\gcd(R, p-1) = R/3.$$

Set $m = R/3$, and retain the full set

$$\mathcal{T}_3 = \{u \mid a^2 : m \mid p + 4u\}.$$

Every member is a rational middle trace whose two tail denominators divide three. Each such trace returns, at unchanged p, R, a , to an original integer E or M state. The exact entire-shell identity is

$$\boxed{2|E| + |M| = |\mathcal{T}_3|}.$$

In particular the number $|E| + |M|/2$ of increasing original triples at this shell equals $|\mathcal{T}_3|/2$.

Proof. Since $3 \mid p-1$, the exact gcd condition implies $9 \mid R$. Therefore $3 \mid m$ and $R \mid m^2$. Also $m > 1$ is odd. Let

$$\omega = (p-1)/m \pmod{3} \in \{1, 2\}.$$

For $u \in \mathcal{T}_3$ put $v = a^2/u$. Modulo m , we have $4(a+u) = p+R+4u = 0$, so $m \mid a+u$. Since $R \mid m^2$,

$$R \mid (a+u)^2.$$

The original middle tail sum is $p(a+u)^2/(Ru)$; it is integral because $u \mid a^2$, $u \mid (a+u)^2$, and $\gcd(u, R) = 1$. Each tail has denominator dividing three.

Dividing the congruence $(a+u)^2 = 0 \pmod{R}$ by the unit u gives

$$u + v + 2a = 0 \pmod{R}.$$

Consequently

$$(p+4u) + (p+4v) = 0 \pmod{R}.$$

It follows that $v \in \mathcal{T}_3$ and that the exact residue

$$b_u = (p+4u)/m \pmod{3}$$

satisfies $b_v = -b_u$. The complementary involution is free here: a fixed point would have $u = a$, and $m \mid p+4a = 2p+R$ would contradict $\gcd(m, 2p) = 1$.

There are now precisely two kinds of complementary pair. If $b_u = 0$, both u, v are full M words and neither is E. If $b_u \neq 0$, the residues b_u, b_v are 1, 2. Exactly one equals ω . Since

$$4u + 1 = (p+4u) - (p-1),$$

that word, and only that word in the pair, is full E; neither is M. All original full words are accounted for: $M \subset \mathcal{T}_3$ immediately, and $u \in E$ gives $p+4u = p-1 \pmod{R}$, so it too belongs to $\mathcal{T}_3$. Counting the two pair types proves the identity.

The output is explicit: use M at u when $b_u = 0$; use E at u when $b_u = \omega$; otherwise use E at v . Recompute the displayed normalization at the chosen divisor. Each full gate proves the corresponding quotient integral and positive. The original reciprocal identity and ordered inverse then give the claimed integer state, without assuming one beforehand. The E order is fixed, and the two M orientations exchange the two tails. Their prime-divisibility patterns distinguish the channels, which proves the stated increasing-triple count. $\square$

Thus the trace-orbit map

$$\mathcal{T}_3/(u \leftrightarrow a^2/u) \longrightarrow E \amalg M/(u \leftrightarrow a^2/u)$$

is a bijection. Its inverse attaches the complementary pair. On a labelled input, retaining the bit indicating the original member restores that member exactly. This is not an equality between a supported zero and an absent word: $b_u = 0$ is an actual original full M word, whereas $\mathcal{T}_3 = \emptyset$ is an empty source.

Corollary 2 (Necessary condition for a complete failure). *At any hypothetical counterexample prime, every original residual satisfying $\gcd(R, p-1) = R/3$ must have $\mathcal{T}_3 = \emptyset$. There cannot be a nonempty third-denominator trace source with both full channels empty at any such shell.*

This strengthens full-gate failure to absence of that larger, explicitly defined rational source. It does not prove that the larger source exists.

Corollary 3 (The entire applicable residual family comes from $p-1$). *Write*

$$p-1 = 2^k 3^\nu b, \quad \gcd(b, 6) = 1, \quad k \geq 3, \quad \nu \geq 1.$$

Every residual satisfying Theorem 1, and no other, is

$$R = 3^{\nu+1}d, \quad d \mid b, \quad d \equiv (-1)^\nu \pmod{4}.$$

All such residuals automatically satisfy $R \leq 3(p-1)/2^k < p$. The family is nonempty precisely when ν is even or b has a prime factor $3 \pmod{4}$. If ν is even, $R = 3^{\nu+1}$ is an unconditional choice of applicable residual for this prescribed prime.

Proof. At the prime 3, the equality $\gcd(R, p-1) = R/3$ forces $v_3(R) = \nu + 1$. At every other prime it requires its exponent in R to be at most its exponent in $p-1$. Since R is odd, this says exactly $R = 3^{\nu+1}d$ with $d \mid b$. The condition $R \equiv 3 \pmod{4}$ is the stated congruence on d ; the bound follows from $d \leq b$. If ν is even, choose $d = 1$. If ν is odd, a divisor $d \equiv 3 \pmod{4}$ exists exactly when some prime factor of b is $3 \pmod{4}$. $\square$

This constructs the whole residual family from an arbitrary prescribed p . It does not assert that any corresponding $\mathcal{T}_3$ is nonempty.

3 An exact factor-level solver at residual 27

Theorem 4 (Full exceptional locus and bounded prime construction). *Let p be prime with $p \equiv 73$ or $145 \pmod{216}$, and put $a = (p+27)/4$. The original residual-27 shell is empty if and only if one of the following disjoint conditions holds:*

1. *Every prime factor of a is $1 \pmod{3}$.*
2. *Exactly two prime-factor occurrences of a are $5 \pmod{9}$, counting multiplicity, and every other prime factor is $1 \pmod{9}$.*

Otherwise an integer E or M state at this shell can be constructed using at most three signed original prime occurrences in u/a , followed by Theorem 1.

Proof. The stated prime classes are exactly the $p \equiv 1 \pmod{24}$ classes with $\gcd(27, p-1) = 9$. Also $a \equiv 7 \pmod{9}$. The source in Theorem 1 is now

$$\mathcal{T}_3 = \{u \mid a^2 : u \equiv 2 \pmod{9}\}.$$

In centred exponents, this is the problem of finding

$$u/a = \prod_q q^{\beta_q} \equiv -1 \pmod{9}, \quad -e_q \leq \beta_q \leq e_q.$$

The cyclic group of units modulo nine is generated by 2. The six residues 1, 2, 4, 8, 7, 5 have logarithms 0, 1, 2, 3, 4, 5 modulo six, respectively. The total logarithm of a is four, and the target is three.

A factor 8 $\pmod{9}$ immediately supplies the target with one exponent. Suppose none occurs. Let N be the total number of prime occurrences in classes 2, 5 $\pmod{9}$. It is even, because the total logarithm is even. If $N \geq 4$, choose any three such occurrences, with exponent +1 for a residue-2 occurrence and -1 for a residue-5 occurrence. Their signed logarithms add to three. Repetition at a prime uses at most its original exponent; every chosen word is inside the original box.

If $N = 2$ and a prime of class 4 or 7 occurs, choose one of the two odd occurrences with sign giving logarithm one. Choose the class-4-or-7 occurrence with sign giving logarithm two. These two exponents

give the target. If $N = 0$, every prime factor is in $\{1, 4, 7\}$ and every available logarithm is even, so the target is absent. This is the first exception.

The only remaining case is $N = 2$, no class 8, and no class 4 or 7. All other prime factors are class 1. The total logarithm four forces both odd occurrences to be class 5, since two values in $\{1, 5\}$ sum to four modulo six only as $5 + 5$. The whole centred support is then contained in $\{-2, -1, 0, 1, 2\} \pmod{6}$, which omits three. This is exactly the second exception, and proves necessity and sufficiency without a residue census.

For each successful construction set $u = \prod q^{e_q + \beta_q}$. All exponents belong to $[0, 2e_q]$ and $u \equiv 2 \pmod{9}$. The preceding theorem decides the full channel and whether to take the actual complementary divisor, then reconstructs the ordered integer denominators. $\square$

4 What this discharges and what it does not

The identity in Theorem 1 is for the complete original shell, not a selected packet, an averaged count, or a post-witness receiver. It removes the simultaneous two-channel cancellation at the stated residuals. Theorem 4 starts from a prescribed prime in its domain and has a complete factor-level exceptional locus and an explicit bounded word construction. Neither proof uses numerical estimates or numerical evidence.

An arbitrary counterexample prime still has to be excluded across all original shells. Even the residual-27 source can be empty. For example, $p = 1009$ gives $a = 259 = 7 \cdot 37$, the first exceptional type, and $p = 73$ gives $a = 25 = 5^2$, the second. These are empty-shell examples, not ES counterexamples. The previously recorded $p = 6841$, $a = 1717 = 17 \cdot 101$ has a residue-8 factor and returns the already known E word $u = 101$, with denominators (1717, 436118, 175499014). It is an illustration of the proved whole-shell map, not a newly discovered prime case.

In particular the universal conclusion

$$\begin{aligned} \forall p \text{ hard prime } \exists R, u : \quad & 0 < R < p, \quad R \equiv 3 \pmod{4}, \\ & u \mid ((p + R)/4)^2, \quad R \mid 4u + 1 \text{ or } R \mid p + 4u. \end{aligned}$$

has not been proved. The remaining task is to force a source in at least one original shell, not to infer one from the lemma's empty cases.

Executable scope and provenance. The standard-library file `arithmetic.py` implements the two exact returns and explicitly reports the exceptional factor types. It tests primality by trial division and returns every original marking and ordered inverse check. The four example records test an E output, an M output, and the two empty-shell types. They are corroboration of the constructors, not proof by a finite range. No numerical scan, historical-priority claim, independent human review, or Lean build is asserted. The result here is a consequence proved from the original divisibility equations; an earlier matching published statement has not been established.

References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. doi:10.1017/S1446788712000468. Primary source: arXiv:1107.1010v6, Section 2, Propositions 2.2 and 2.6. The classical parametrization is used, not a pointwise existence theorem.